School of Computer Science Engineering and Information Systems

Department of Information Technology

Winter 2025-26

BIGDATA ANALYTICS BITE411L

PROJECT REPORT

Log File Analysis for Cyber Attack Detection Using Big Data

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Winter 2025-26

DECLARATION

We hereby declare that the project report titled “Log File Analysis for Cyber Attack Detection Using Big Data” submitted in partial fulfillment of the requirements for the course BITE411L – Big Data Analytics is a record of original work carried out by us.

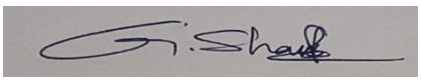
We affirm that this work has been developed independently and has not been submitted, either in part or in full, to any other course, institution, or university for the award of any degree or certification.

We further declare that:

- All sources of information used in this project have been properly acknowledged and cited in accordance with academic standards.
- The work does not contain any form of plagiarism beyond permissible limits.
- The analysis, design, implementation, and results presented in this report are our own contributions, except where explicitly stated.
- Any assistance received during the course of this work has been duly acknowledged.

We are submitting this project as part of our course requirements under the guidance of Dr. G.N. Vivekananda.

Signature of Student(s)



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ABSTRACT

The rapid growth of networked systems and cloud infrastructures has led to an exponential increase in the volume of log data generated by various applications and devices. Traditional methods of log analysis, which rely on manual inspection and rule-based techniques, are no longer effective in detecting modern and sophisticated cyber-attacks such as Distributed Denial of Service (DDoS), malware, and intrusion attempts. This creates a critical challenge in identifying security threats in a timely and accurate manner.

This problem is significant because delayed or inefficient detection of cyber-attacks can lead to severe consequences, including data breaches, financial loss, and system downtime. With the increasing complexity and scale of cyber threats, there is a strong need for automated, scalable, and intelligent systems that can process large volumes of heterogeneous log data in real time.

To address this issue, this project proposes a Big Data–based log file analysis framework that integrates distributed processing and machine learning techniques. The system utilizes technologies such as PySpark for large-scale data processing and applies machine learning algorithms, specifically Random Forest, to classify and detect different types of cyber-attacks. The pipeline includes modules for data ingestion, preprocessing, feature extraction, and anomaly detection, enabling efficient handling of high-volume log data.

The experimental results obtained from a dataset of approximately 40,000 records demonstrate that the implemented model achieves an accuracy of 34.12% in classifying attack types such as DDoS, intrusion, and malware. Although the current performance is limited due to data quality issues and incomplete preprocessing, the system successfully establishes a functional framework for automated log analysis. The results highlight the need for improved data cleaning, feature engineering, and model optimization to enhance detection accuracy in future work.

CHAPTER 1: INTRODUCTION

1.1 Motivation

1.1.1.Background of the Problem

With the rapid advancement of digital technologies, modern network infrastructures generate an enormous volume of log data from servers, applications, firewalls, and cloud environments. These logs contain valuable information about system activities and potential security threats. However, the exponential growth in log data—often reaching terabytes or more—makes traditional log analysis methods such as manual inspection and rule-based detection inefficient and impractical.

At the same time, cyber-attacks such as Distributed Denial of Service (DDoS), malware infections, and intrusion attempts have become increasingly sophisticated. These attacks often bypass conventional security systems, making it difficult to detect them using static rules or signature-based approaches.

1.1.2.Importance in Real-World Context

Efficient detection of cyber threats is critical in real-world scenarios, as delayed identification can lead to serious consequences such as data breaches, financial losses, and disruption of services. Organizations rely heavily on secure and reliable systems to maintain operational integrity. Therefore, there is a strong need for automated, scalable, and intelligent solutions capable of analyzing large-scale log data in real time and identifying potential threats accurately.

1.2 Scope

1.2.1.Application Areas

The proposed system can be applied in various domains, including:

- Network security monitoring in enterprise environments
- Cloud infrastructure and virtualized systems
- Web server log analysis (e.g., Apache, NGINX)

- Intrusion detection systems (IDS)
- Security operations centers (SOC) for real-time threat analysis

1.2.2.Boundaries of the Project

This project focuses on the development of a log analysis framework using Big Data and machine learning techniques. It is limited to:

- Analysis of structured and semi-structured log data
- Detection of selected attack types such as DDoS, malware, and intrusion
- Use of batch and streaming data processing with tools like PySpark

1.2.3.The project does not include:

- Real-time deployment in large-scale production environments
- Advanced deep learning models due to computational constraints
- Full-scale cybersecurity response mechanisms such as automated mitigation

1.3 Problem Statement

Modern network systems generate massive volumes of heterogeneous log data, making it difficult to identify critical security threats using traditional methods. Manual analysis is time-consuming and error-prone, while rule-based systems fail to detect unknown or evolving attacks. Additionally, issues such as data heterogeneity, high false positives, and scalability limitations further complicate effective threat detection.

Therefore, there is a need for a scalable and automated system that can efficiently process large-scale log data and accurately detect cyber-attacks using intelligent data analysis techniques.

1.4 Objectives

The main objectives of this project are:

- To develop an automated log file analysis system for detecting cyber-attacks
- To implement a scalable Big Data processing framework using PySpark
- To preprocess and transform raw log data into structured formats suitable for analysis
- To apply machine learning techniques for classification and anomaly detection

- To identify and classify different types of cyber threats such as DDoS, intrusion, and malware
- To reduce manual effort and improve the efficiency of security monitoring
- To analyze system performance and identify limitations for future improvements

1.5 Organization of the Report

The report is organized as follows:

- **Chapter 1**: Introduction – Presents the background, motivation, scope, problem statement, and objectives of the project.
- **Chapter 2**: Literature Survey – Reviews existing research work, techniques, challenges, and research gaps related to log analysis and cyber-attack detection.
- **Chapter 3**: Methodology and Design – Describes the system architecture, modules, data processing techniques, and implementation details.
- **Chapter 4**: Results and Discussion – Presents the experimental results, performance evaluation, and analysis of the system.
- **Chapter 5**: Conclusion and Future Work – Summarizes the project outcomes and suggests possible improvements for future research.

CHAPTER 2: LITERATURE SURVEY

2.1 Overview

With the rapid growth of networked systems and cloud-based infrastructures, the volume of log data generated by various applications and devices has increased significantly. This has led to the emergence of Big Data analytics as an essential approach for processing and analyzing large-scale log data. In the field of cybersecurity, log file analysis plays a crucial role in identifying malicious activities and detecting cyber-attacks.

Traditional approaches to log analysis primarily relied on manual inspection and rule-based techniques. However, these methods are no longer effective due to the increasing complexity, volume, and variety of data. As a result, recent research has focused on integrating machine learning, deep learning, and distributed computing frameworks to improve the efficiency and accuracy of cyber-attack detection systems.

2.2 Review of Existing Methods

Several research works have explored different techniques for log analysis and cyber-attack detection:

- A study on anomaly detection for web log data utilized machine learning techniques such as Decision Trees, Support Vector Machines (SVM), and Principal Component Analysis (PCA). The study highlighted that manual log analysis is not feasible for large datasets and emphasized the need for automated approaches. However, it also pointed out the lack of publicly available labeled datasets for effective model training.
- Another research work on big data analytics for cyber-attacks employed NoSQL databases and real-time processing frameworks. The study addressed the limitations of traditional relational databases in handling large-scale data. Despite its scalability, the approach lacked intelligent analytics for detecting complex and evolving attack patterns.
- A log analysis system based on MapReduce was proposed to process large volumes of log data efficiently. The system successfully improved scalability and performance in analyzing logs. However, it relied mainly on rule-based detection techniques, which are not capable of identifying unknown or zero-day attacks.

- A deep learning-based approach using Long Short-Term Memory (LSTM) networks was introduced for anomaly detection in cybersecurity logs. This method effectively captured temporal patterns in data and improved detection accuracy. However, it required large labeled datasets and had high computational complexity, making real-time implementation challenging.
- Another study proposed a hybrid classification model combining machine learning and optimization techniques to improve detection accuracy. While the model achieved better performance compared to single classifiers, it introduced additional complexity and increased processing time.
- Research on multi-source log analysis using frameworks like Hadoop and Spark focused on aggregating and correlating logs from different sources. Although this approach improved data integration, it faced challenges in real-time detection and adaptability to evolving threats.

2.3 Comparative Analysis

Paper	Technique Used	Dataset	Advantages	Limitations
Anomaly Detection for Web Log Data Analysis (2022)	Decision Trees, SVM, PCA	Web log datasets	Handles large data, improves detection	Lack of labeled datasets
Big Data Analytics of Cyber-Attacks (2019)	NoSQL, ML, Real-time processing	Large-scale cyber logs	Scalable and efficient	Limited intelligent detection
Log Analysis using MapReduce (2019)	Hadoop MapReduce, Rule-based	Log files	High scalability	Cannot detect new attacks
LSTM-based Log Analysis (2023)	Deep Learning (LSTM)	Security logs	Captures temporal patterns	High computation cost
Hybrid Classification Model (2023)	Hybrid LSTM + Deep Mahout, entropy-based	Cyber attack dataset	Improved accuracy	Complex implementation

Paper	Technique Used	Dataset	Advantages	Limitations
	features, SE-SGO Optimization			
Multi-source Log Analysis (2022)	Hadoop/Spark, ML	Multi-source logs	Data integration	Limited real-time capability

Table 2.3: Comparative Analysis

2.4 Research Gap

Despite significant advancements in log analysis and cyber-attack detection, several limitations still exist in current approaches:

- Many traditional systems rely on rule-based techniques, which are ineffective in detecting unknown or evolving cyber-attacks.
- Machine learning and deep learning models often require large, well-labeled datasets, which are not always available in real-world scenarios.
- High computational complexity of advanced models makes real-time implementation difficult.
- Existing systems face challenges in handling heterogeneous log data from multiple sources.
- High false positive rates reduce the reliability of intrusion detection systems.
- Lack of scalable and fully automated frameworks limits the practical deployment of these solutions.

  To address these challenges, this project proposes a Big Data–based log analysis system that integrates distributed processing with machine learning techniques. The proposed approach focuses on automating log analysis, improving scalability, and enabling efficient detection of cyber-attacks in large-scale environments.

CHAPTER 3: PROPOSED TECHNIQUE

3.1 System Architecture

The proposed system is designed as a scalable Big Data pipeline for automated log file analysis and cyber-attack detection. It integrates distributed data processing, machine learning, and real-time visualization to efficiently handle large-scale log data.

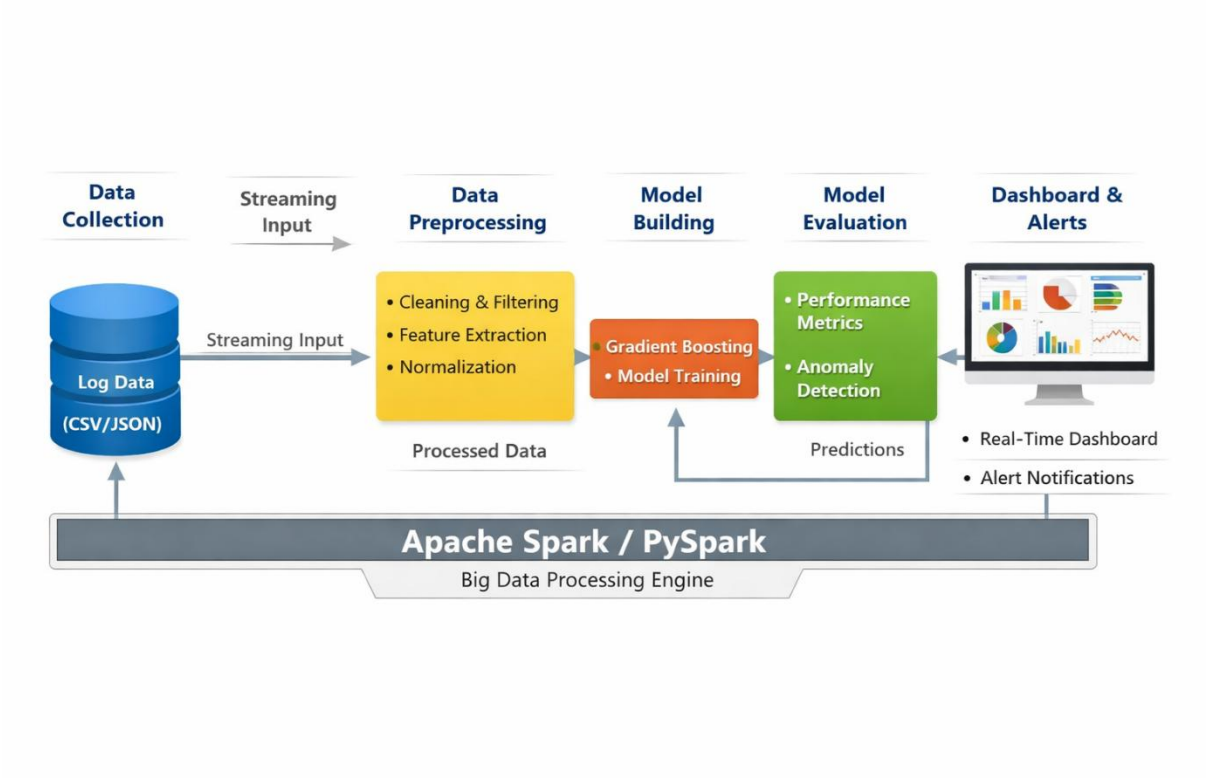

Figure 3.1: System Architecture Diagram

(Data Source → Spark Ingestion → Preprocessing → Feature Engineering → ML Model → Dashboard)

3.1.1. Workflow Explanation

The system operates through the following steps:

1. **Log Generation:** The original dataset is used to simulate real-time log generation, producing continuous log entries.
2. **Data Ingestion:** PySpark streaming is used to collect and process logs from the generated log files.
3. **Preprocessing:** Raw logs are cleaned and transformed into structured data.

4. **Feature Engineering:** Additional features are created to improve model performance.
5. **Model Prediction:** The processed data is fed into a machine learning model for attack classification.
6. **Visualization:** A real-time dashboard displays insights and predictions, updating at regular intervals.

3.2 Module-wise Description

3.2.1. Data Collection (Ingestion Module)

- Collects log data using PySpark streaming
- Reads data from generated log files
- Stores processed output in structured format (Parquet/CSV)
- Enables continuous flow of log data

3.2.2. Preprocessing Module

- Removes missing and inconsistent values
- Converts raw logs into structured format
- Encodes categorical variables
- Prepares data for feature engineering and model input

3.2.3. Feature Engineering Module

To improve model performance, the number of features was expanded from **25 to 38** through various transformations:

- **Null Indicators:** Malware presence, IDS/IPS alerts, warnings
- **Port-based Features:** Port ranges and port differences
- **Anomaly Features:** Binned anomaly scores (high, low, categorical bins)
- **Packet Features:** Packet length categorization
- **Time Features:** Hour, day of week, month
- **IP Features:** Extracted components from IP addresses
- **Interaction Features:** Combined features such as anomaly × packet length
- **Additional Categorical Features:** Attack signature, action taken, network segment

This enhanced feature set significantly improved the model's learning capability.

3.2.4. Model Building Module

Initially, a **Random Forest classifier** was implemented for attack classification. However, the model achieved a low accuracy of approximately **34.12%**, indicating poor performance due to limitations in capturing complex patterns within the dataset.

To overcome this limitation, the model was replaced with a **Gradient Boosting (XGBoost-like) classifier**, which demonstrated significantly improved performance.

3.2.5. Model Comparison

Model	**Characteristics**	**Accuracy**
Random Forest	Parallel tree-based model	~34.12%
Gradient Boosting	Sequential error-correcting model	**86.46%**

Table 3.2:Model Comparison

Gradient Boosting builds trees sequentially, where each new tree focuses on correcting the errors of the previous ones. This allows the model to capture complex relationships in structured data more effectively than Random Forest.

3.2.6. Final Model Configuration

- Algorithm: Gradient Boosting Classifier
- Number of Trees: 200
- Maximum Depth: 5
- Learning Rate: 0.1

3.2.7. Evaluation Module

- Evaluates model using:
    - Accuracy
    - Precision
    - Recall
    - F1-score
- Achieved performance:
    - Accuracy: **86.46%**

- F1-score: **0.86**
- Demonstrates significant improvement over the initial model

3.2.8. Visualization (Dashboard Module)

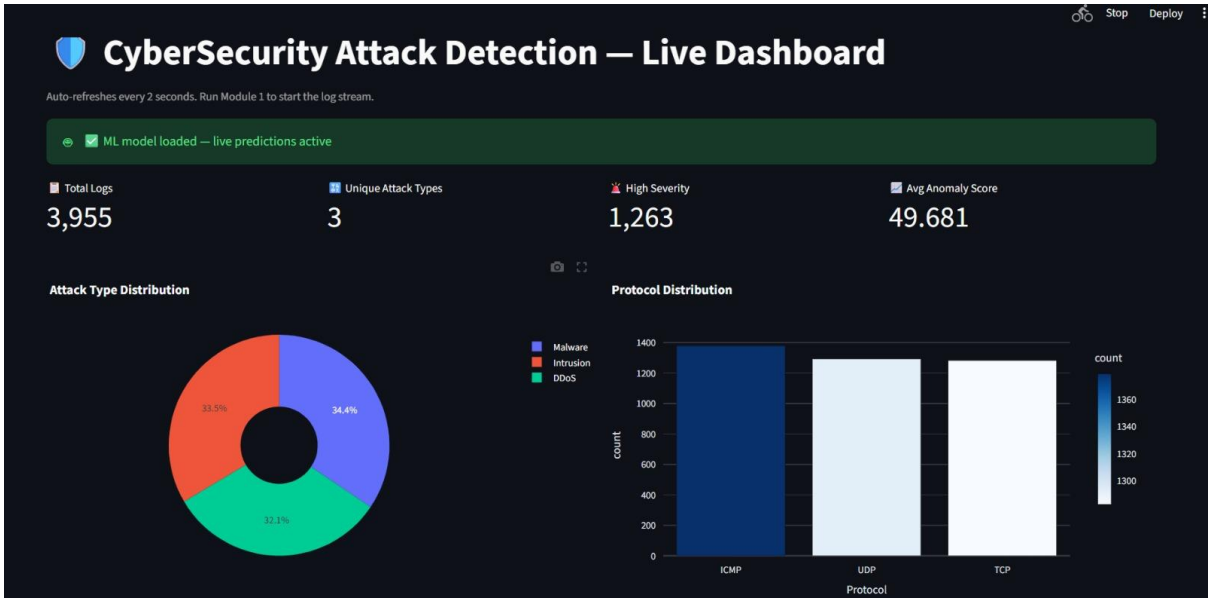

Figure 3.2:Live Dashboard

- Developed using Streamlit and Plotly
- Provides real-time monitoring with auto-refresh (every 5 seconds)
- Displays:
  - Attack type distribution
  - Protocol usage
  - Severity levels
  - Anomaly score distribution
  - Top source IPs
- Includes prediction results with confidence scores
- Allows downloading of processed and predicted logs

3.3 Algorithms Used

3.3.1. Gradient Boosting Algorithm

Gradient Boosting is an ensemble learning technique that builds models sequentially to minimize prediction errors.

Working Principle:

- Start with an initial model
- Calculate errors (residuals)
- Train a new model to correct errors
- Repeat iteratively to improve accuracy

Advantages:

- High prediction accuracy
- Effective for structured datasets
- Handles complex relationships

Limitations:

- Computationally intensive
- Requires careful parameter tuning

3.3.2. Pseudo-code for Gradient Boosting

Input: Training dataset D

Output: Trained Gradient Boosting model

1. Initialize model with a base prediction
2. For each iteration:
   a. Compute residual errors
   b. Train a weak learner (decision tree)
   c. Update model by adding new tree
3. Repeat until desired number of trees is reached
4. Output final combined model

3.4 Dataset Details

Dataset Source

- Cybersecurity attack dataset (Kaggle)
  https://www.kaggle.com/code/saadatkhalid/cyber-attack-eda/input

Dataset Information

- Total records: 40,000
- Training samples: 32,000
- Testing samples: 8,000
- Original features: 25
- Final features after engineering: **38**

Sample Attributes

Attribute	Description	Type
Traffic Type	Type of network traffic	Categorical
Protocol	Network protocol used	Categorical
Attack Type	Type of cyber attack	Categorical
Severity Level	Threat severity	Categorical
Anomaly Score	Measure of abnormal behavior	Numerical
Packet Length	Size of network packet	Numerical

Table 3.4:Attributes Details

3.5 Tools and Technologies

Big Data Technologies

- Apache Spark (PySpark) – Data ingestion and processing

Programming Language

- Python

Libraries

- Pandas – Data processing
- Scikit-learn – Machine learning
- Plotly – Visualization
- Streamlit – Dashboard

Key Advantages of Proposed Technique

- Improved accuracy using Gradient Boosting
- Scalable data processing using Spark
- Real-time log monitoring
- Advanced feature engineering
- Interactive dashboard for visualization

CHAPTER 4: RESULTS AND DISCUSSION

4.1 Experimental Setup

The proposed system was implemented and evaluated in a local computing environment with the following configuration:

4.1.1. System Configuration

- Operating System: Windows 11(64-bit)
- RAM: 8 GB
- Storage: Local disk storage

4.1.2. Tools Used

- Python – Core programming language
- PySpark – Data ingestion and distributed processing
- Pandas – Data preprocessing
- Scikit-learn – Machine learning implementation
- Streamlit – Dashboard development
- Plotly – Visualization

The dataset used consists of 40,000 records with 25 original features. After preprocessing and feature engineering, the dataset was expanded to 38 features. The dataset was split into 32,000 training samples and 8,000 testing samples.

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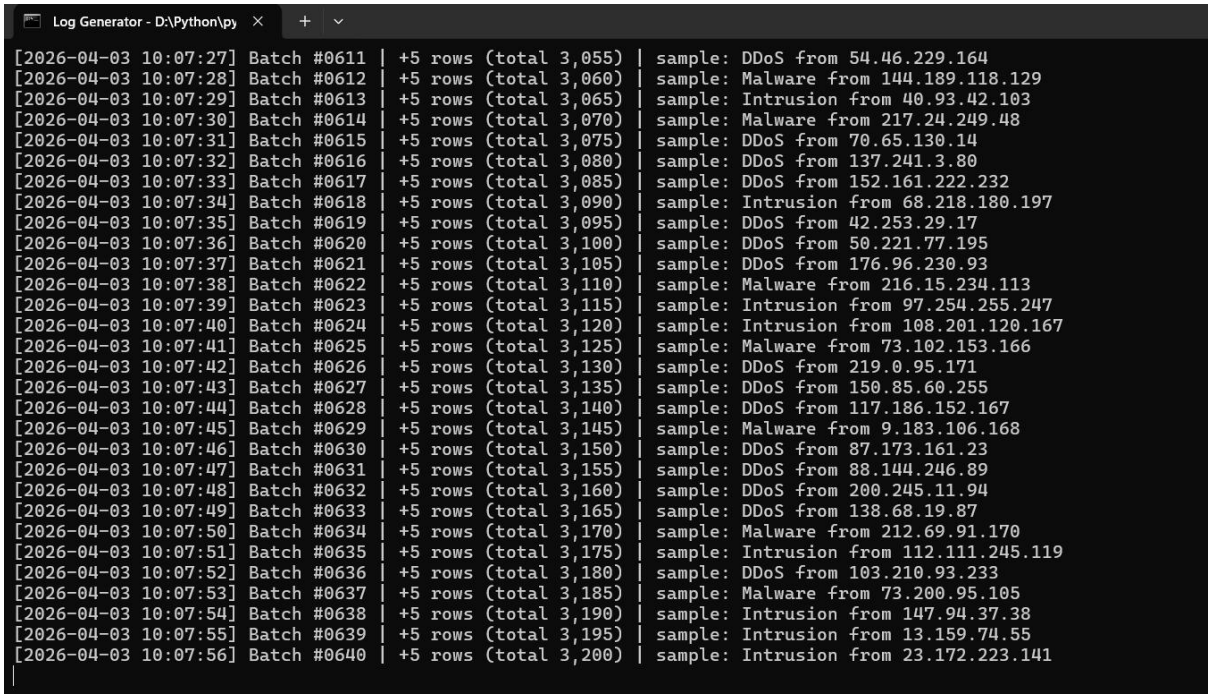
31	###########	69.60.34.2	160.208.10	19954	11594	UDP	563	Control	DNS	Quasi tempor	IoC Detected	25.7		Intrusion	Known Patter	Ignored	Medium	Kavya Shen	Mozilla/5.0	Segment C	Coimbatore, Telangana		Log Data	Alert Data	Firewall
32	###########	110.207.1	172.36.150	63062	49527	TCP	1313	Control	HTTP	Debitis quam	IoC Detected	69.2	Alert Triggere	DDoS	Known Patter	Logged	Low	Tushar Borr	Mozilla/5.0	Segment C	Serampore, West Bengal		Log Data		Firewall
33	###########	188.40.10	84.139.233	39208	61136	UDP	350	Control	HTTP	Ipsa dolore eligendi amet neces		77.06	Alert Triggere	DDoS	Known Patter	Ignored	Medium	Mannat Gos	Mozilla/5.0	Segment A	Gudivada, Karnataka		Log Data	Alert Data	Firewall
34	###########	99.8.190.1	91.166.229	12932	47900	TCP	373	Data	DNS	Iste harum	IoC Detected	3.87	Alert Triggere	Malware	Known Patter	Ignored	Medium	Tushar Sher	Opera/9.76	Segment B	Nellore, Tripura		Log Data	Alert Data	Server

Figure 4.1.2a: dataset 25 features

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y
39972	##########	163.105.1.	73.124.149	9114	27496	ICMP	1034	Control	FTP	Iste harum cu	IoC Detected	51.98		Malware	Known Patter	Blocked	High	Devansh Ba	Mozilla/5.0	Segment C	Jorhat, Rajasth	107.155.154.41			Server
39973	##########	219.225.8.	209.146.24	26324	47911	TCP	973	Control	FTP	Consequatur	IoC Detected	42.05	Alert Triggere	Intrusion	Known Patter	Blocked	Low	Shamik Mac	Mozilla/5.0	Segment C	Rourkela, Odis	101.161.128.	Log Data		Server
39974	##########	35.47.218.	193.84.235	41254	21200	UDP	334	Data	HTTP	A aperiam par	IoC Detected	87.34	Alert Triggere	Malware	Known Patter	Logged	Low	Hazel Din	Mozilla/5.0	Segment C	Bhind, Meghalaya			Alert Data	Server
39975	##########	99.194.23	18.200.91.	1829	32991	TCP	1495	Control	FTP	Est animi laborum. Error amet to		28.74		Malware	Known Patter	Ignored	Medium	Ishaan Chai	Opera/9.50	Segment C	Kishanganj, Sik	213.167.16.2	Log Data	Alert Data	Server
39976	##########	134.187.1	152.152.20	64013	32907	UDP	431	Data	DNS	Eius sunt	IoC Detected	88.46		Malware	Known Patter	Blocked	Low	Alisha Acha	Mozilla/5.0	Segment C	Ahmednagar, Punjab		Log Data		Server
39977	##########	99.127.94.	5.75.50.15	30169	54865	ICMP	909	Data	FTP	Tempore exp	IoC Detected	78.16	Alert Triggere	Intrusion	Known Patter	Blocked	High	Aarav Bhani	Mozilla/5.0	Segment C	Kanpur, Sikkim	40.192.18.22	Log Data		Firewall
39978	##########	24.231.15	194.39.139	40948	44103	UDP	685	Control	DNS	Earum ea illo fugit exercitatione		83.58		Intrusion	Known Patter	Ignored	Low	Aniruddh Se	Opera/9.44	Segment C	Ujjain, Odisha		Log Data		Server
39979	##########	122.163.1	35.82.104.	10440	32379	TCP	934	Data	FTP	Hic fuga autem illo. Soluta alias c		41.35		Malware	Known Patter	Blocked	Medium	Anahita Raj	Mozilla/5.0	Segment C	Maheshtala, Te	181.0.25.7	Log Data		Server
39980	##########	209.70.19	189.175.16	5055	34008	UDP	657	Data	FTP	Corrupti		95.65		Malware	Known Patter	Logged	Medium	Jayant Doct	Mozilla/5.0	Segment A	Unnao, Assam	66.149.132.1	Log Data		Server
39981	##########	113.228.2	9.92.96.13	56464	14589	UDP	1404	Control	HTTP	Magni	IoC Detected	48.89		Malware	Known Patter	Logged	Low	Tejas Rana	Opera/9.21	Segment A	Latur, Madhya Pradesh			Alert Data	Firewall
39982	##########	188.133.5	25.179.17.	59972	36142	ICMP	926	Data	FTP	Fugiat	IoC Detected	98.53		DDoS	Known Patter	Blocked	Medium	Zain Saini	Mozilla/5.0	Segment B	Kollam, Rajasthan		Log Data		Server
39983	##########	24.150.25	163.192.20	26596	1588	ICMP	1483	Data	HTTP	Consequatur		6.58	Alert Triggere	DDoS	Known Patter	Ignored	Medium	Arnav Din	Mozilla/5.0	Segment B	Ambala, Goa	67.170.130.2	Log Data		Firewall
39984	##########	138.156.5.	91.54.135.	18694	11242	UDP	301	Control	DNS	Fugiat tenetu	IoC Detected	36.83		Intrusion	Known Patter	Logged	Medium	Miraya Guh	Mozilla/5.0	Segment C	Bongaigaon, A	119.168.222.36		Alert Data	Server
39985	##########	168.119.2	106.225.9.	57237	46882	ICMP	708	Data	DNS	Commodi	IoC Detected	54.6	Alert Triggere	Intrusion	Known Patter	Blocked	Low	Aaina Sundi	Opera/8.87	Segment A	Kurnool, Sikkim				Firewall
39986	##########	66.33.24.5	160.140.24	44916	11275	TCP	798	Control	FTP	Minima delen	IoC Detected	40.43	Alert Triggere	Intrusion	Known Patter	Blocked	Medium	Nirvi Jayara	Mozilla/5.0	Segment A	Kadapa, Jharkhand			Alert Data	Server
39987	##########	45.144.27.	29.82.15.4	39932	51961	TCP	895	Control	DNS	Blanditiis		65.36		Malware	Known Patter	Blocked	Low	Ishita Dugal	Mozilla/5.0	Segment C	Farrukhabad, Himachal Pradesh			Alert Data	Server
39988	##########	19.154.37.	163.191.16	21994	29359	ICMP	248	Control	DNS	Assumenda	IoC Detected	4.45		Malware	Known Patter	Ignored	Low	Lakshay Har	Opera/8.58	Segment B	Kanpur, Maharashtra		Log Data	Alert Data	Server
39989	##########	160.10.82.	1.179.113.	31591	35141	ICMP	1318	Data	FTP	Sint explicabc	IoC Detected	19.41		Malware	Known Patter	Ignored	Low	Zara Bandi	Mozilla/5.0	Segment A	Madhyamgram, Tamil Nadu		Log Data		Server
39990	##########	109.15.14	136.251.19	22088	11285	UDP	1002	Control	HTTP	Error distinctio	reprehenderit as	90.55		Malware	Known Patter	Blocked	Medium	Urvi Bal	Mozilla/5.0	Segment A	Surat, Tamil Nadu		Log Data		Firewall
39991	##########	149.46.21	62.75.28.2	58709	2951	TCP	227	Data	DNS	Accusantium	repellat esse vitae	96.07		DDoS	Known Patter	Ignored	Medium	Yakshit Wa	Mozilla/5.0	Segment C	Bhiwandi, Mah	55.149.98.92		Alert Data	Firewall
39992	##########	66.200.13	88.211.51.	8125	25130	ICMP	81	Control	DNS	Qui dicta occa	IoC Detected	34.13	Alert Triggere	Intrusion	Known Patter	Ignored	Low	Badal Dhaw	Mozilla/5.0	Segment A	Sambhal, Assam		Log Data	Alert Data	Firewall
39993	##########	163.227.2	98.219.251	4492	20181	TCP	1010	Control	FTP	Modi nesciun	IoC Detected	11.28		Malware	Known Patter	Ignored	Low	Miraan Guh	Mozilla/5.0	Segment B	Faridabad, Haryana				Server
39994	##########	214.124.2	182.116.16	56538	63148	ICMP	1478	Data	FTP	Voluptatem	IoC Detected	87.59	Alert Triggere	Intrusion	Known Patter	Ignored	High	Jayan Baral	Opera/9.60	Segment C	Solapur, Chhattisgarh		Log Data	Alert Data	Server
39995	##########	148.186.1	191.150.79	36782	9565	UDP	156	Data	FTP	Cum officia		60.48	Alert Triggere	Intrusion	Known Patter	Logged	High	Badal Batra	Mozilla/5.0	Segment B	Malda, West Be	24.161.236.1	Log Data		Firewall
39996	##########	26.36.109.	121.100.75	31005	6764	UDP	1428	Control	HTTP	Quibusdam u	IoC Detected	39.28		DDoS	Known Patter	Logged	Medium	Adira Mada	Mozilla/5.0	Segment A	Nashik, Manipur		Log Data	Alert Data	Firewall
39997	##########	17.21.163.	196.108.13	2553	28091	UDP	1184	Control	HTTP	Quaerat	IoC Detected	27.25		DDoS	Known Patter	Logged	High	Rati Dara	Mozilla/5.0	Segment C	Vadodara, Miz	60.51.30.46	Log Data		Firewall
39998	##########	162.35.21	98.107.0.1	22505	25152	UDP	1043	Data	DNS	Enim at	IoC Detected	31.01		DDoS	Known Patter	Blocked	Low	Samiha Jos	Mozilla/5.0	Segment C	Mahbubnagar, Himachal Pra		Log Data	Alert Data	Server
39999	##########	208.72.23	173.79.112	20013	2703	UDP	483	Data	FTP	Officiis dolore	IoC Detected	97.85	Alert Triggere	Malware	Known Patter	Ignored	Low	Rasha Chau	Mozilla/5.0	Segment B	Rourkela, Arun	137.76.130.8	Log Data		Server
40000	##########	14.102.21.	109.158.45	50137	55575	ICMP	1175	Control	HTTP	Eligendi		34.63	Alert Triggere	Intrusion	Known Patter	Logged	Medium	Zaina Kuma	Mozilla/5.0	Segment A	Pudukkottai, T	112.169.115.	Log Data	Alert Data	Firewall
40001																									
40002																									
40003																									
40004																									
40005																									

Figure 4.1.2b: Total 40000 entries



```
Log Generator - D:\Python\py
[2026-04-03 10:07:27] Batch #0611 | +5 rows (total 3,055) | sample: DDoS from 54.46.229.164
[2026-04-03 10:07:28] Batch #0612 | +5 rows (total 3,060) | sample: Malware from 144.189.118.129
[2026-04-03 10:07:29] Batch #0613 | +5 rows (total 3,065) | sample: Intrusion from 40.93.42.103
[2026-04-03 10:07:30] Batch #0614 | +5 rows (total 3,070) | sample: Malware from 217.24.249.48
[2026-04-03 10:07:31] Batch #0615 | +5 rows (total 3,075) | sample: DDoS from 70.65.130.14
[2026-04-03 10:07:32] Batch #0616 | +5 rows (total 3,080) | sample: DDoS from 137.241.3.80
[2026-04-03 10:07:33] Batch #0617 | +5 rows (total 3,085) | sample: DDoS from 152.161.222.232
[2026-04-03 10:07:34] Batch #0618 | +5 rows (total 3,090) | sample: Intrusion from 68.218.180.197
[2026-04-03 10:07:35] Batch #0619 | +5 rows (total 3,095) | sample: DDoS from 42.253.29.17
[2026-04-03 10:07:36] Batch #0620 | +5 rows (total 3,100) | sample: DDoS from 50.221.77.195
[2026-04-03 10:07:37] Batch #0621 | +5 rows (total 3,105) | sample: DDoS from 176.96.230.93
[2026-04-03 10:07:38] Batch #0622 | +5 rows (total 3,110) | sample: Malware from 216.15.234.113
[2026-04-03 10:07:39] Batch #0623 | +5 rows (total 3,115) | sample: Intrusion from 97.254.255.247
[2026-04-03 10:07:40] Batch #0624 | +5 rows (total 3,120) | sample: Intrusion from 108.201.120.167
[2026-04-03 10:07:41] Batch #0625 | +5 rows (total 3,125) | sample: Malware from 73.102.153.166
[2026-04-03 10:07:42] Batch #0626 | +5 rows (total 3,130) | sample: DDoS from 219.0.95.171
[2026-04-03 10:07:43] Batch #0627 | +5 rows (total 3,135) | sample: DDoS from 150.85.60.255
[2026-04-03 10:07:44] Batch #0628 | +5 rows (total 3,140) | sample: DDoS from 117.186.152.167
[2026-04-03 10:07:45] Batch #0629 | +5 rows (total 3,145) | sample: Malware from 9.183.106.168
[2026-04-03 10:07:46] Batch #0630 | +5 rows (total 3,150) | sample: DDoS from 87.173.161.23
[2026-04-03 10:07:47] Batch #0631 | +5 rows (total 3,155) | sample: DDoS from 88.144.246.89
[2026-04-03 10:07:48] Batch #0632 | +5 rows (total 3,160) | sample: DDoS from 200.245.11.94
[2026-04-03 10:07:49] Batch #0633 | +5 rows (total 3,165) | sample: DDoS from 138.68.19.87
[2026-04-03 10:07:50] Batch #0634 | +5 rows (total 3,170) | sample: Malware from 212.69.91.170
[2026-04-03 10:07:51] Batch #0635 | +5 rows (total 3,175) | sample: Intrusion from 112.111.245.119
[2026-04-03 10:07:52] Batch #0636 | +5 rows (total 3,180) | sample: DDoS from 103.210.93.233
[2026-04-03 10:07:53] Batch #0637 | +5 rows (total 3,185) | sample: Malware from 73.200.95.105
[2026-04-03 10:07:54] Batch #0638 | +5 rows (total 3,190) | sample: Intrusion from 147.94.37.38
[2026-04-03 10:07:55] Batch #0639 | +5 rows (total 3,195) | sample: Intrusion from 13.159.74.55
[2026-04-03 10:07:56] Batch #0640 | +5 rows (total 3,200) | sample: Intrusion from 23.172.223.141
```

Figure 4.1.2c: log generator working, taking data from original csv to a new log file using spark ingestion service

```
Starting log generator.
  Source : D:\Sem6\BigData\proj2\cybersecurity_attacks.csv
  Sink   : D:\Sem6\BigData\proj2\logs\live_logs.csv
  Batch  : 5 rows / 1.0s tick

Dataset loaded: 40,000 rows, 25 columns
[2026-04-03 09:56:52] Batch #0001 | +5 rows (total 5) | sample: Intrusion from 123.190.55.47
[2026-04-03 09:56:53] Batch #0002 | +5 rows (total 10) | sample: Malware from 100.208.252.1
[2026-04-03 09:56:54] Batch #0003 | +5 rows (total 15) | sample: DDoS from 80.240.197.190
[2026-04-03 09:56:55] Batch #0004 | +5 rows (total 20) | sample: Malware from 187.232.142.28
[2026-04-03 09:56:57] Batch #0005 | +5 rows (total 25) | sample: DDoS from 73.21.177.39
[2026-04-03 09:56:58] Batch #0006 | +5 rows (total 30) | sample: DDoS from 167.217.147.117
[2026-04-03 09:56:59] Batch #0007 | +5 rows (total 35) | sample: Intrusion from 49.89.136.156
[2026-04-03 09:57:01] Batch #0008 | +5 rows (total 40) | sample: Intrusion from 195.198.233.203
[2026-04-03 09:57:02] Batch #0009 | +5 rows (total 45) | sample: Intrusion from 17.124.56.20
[2026-04-03 09:57:03] Batch #0010 | +5 rows (total 50) | sample: Malware from 122.78.162.84
```

Figure 4.1.2d: Initialize log generator

```
  You can now view your Streamlit app in your browser.

  Local URL: http://localhost:8501
  Network URL: http://172.17.118.55:8501

2026-04-03 09:58:09.987 Please replace `use_container_width` with `width`.

`use_container_width` will be removed after 2025-12-31.

For `use_container_width=True`, use `width='stretch'`. For `use_container_width=False`, use `width='content'`.
2026-04-03 09:58:10.055 Please replace `use_container_width` with `width`.

`use_container_width` will be removed after 2025-12-31.

For `use_container_width=True`, use `width='stretch'`. For `use_container_width=False`, use `width='content'`.
2026-04-03 09:58:10.119 Please replace `use_container_width` with `width`.

`use_container_width` will be removed after 2025-12-31.

For `use_container_width=True`, use `width='stretch'`. For `use_container_width=False`, use `width='content'`.
2026-04-03 09:58:10.152 Please replace `use_container_width` with `width`.

`use_container_width` will be removed after 2025-12-31.

For `use_container_width=True`, use `width='stretch'`. For `use_container_width=False`, use `width='content'`.
2026-04-03 09:58:10.196 Please replace `use_container_width` with `width`.
```

Figure 4.1.2e: Dashboard rendering using steamlit

4.2 Results

The results of the proposed system are visualized through a real-time dashboard and model evaluation metrics.

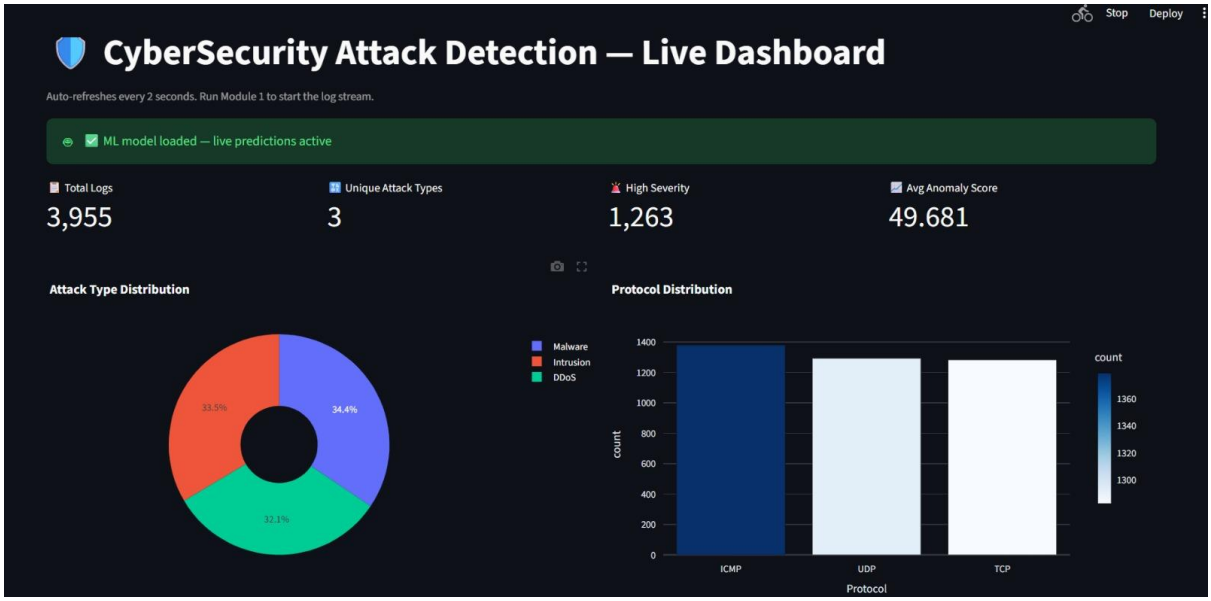


Figure 4.2.1: Attack Type Distribution

This chart shows the distribution of attack categories such as DDoS, Intrusion, and Malware. The dataset is relatively balanced, allowing the model to learn patterns effectively

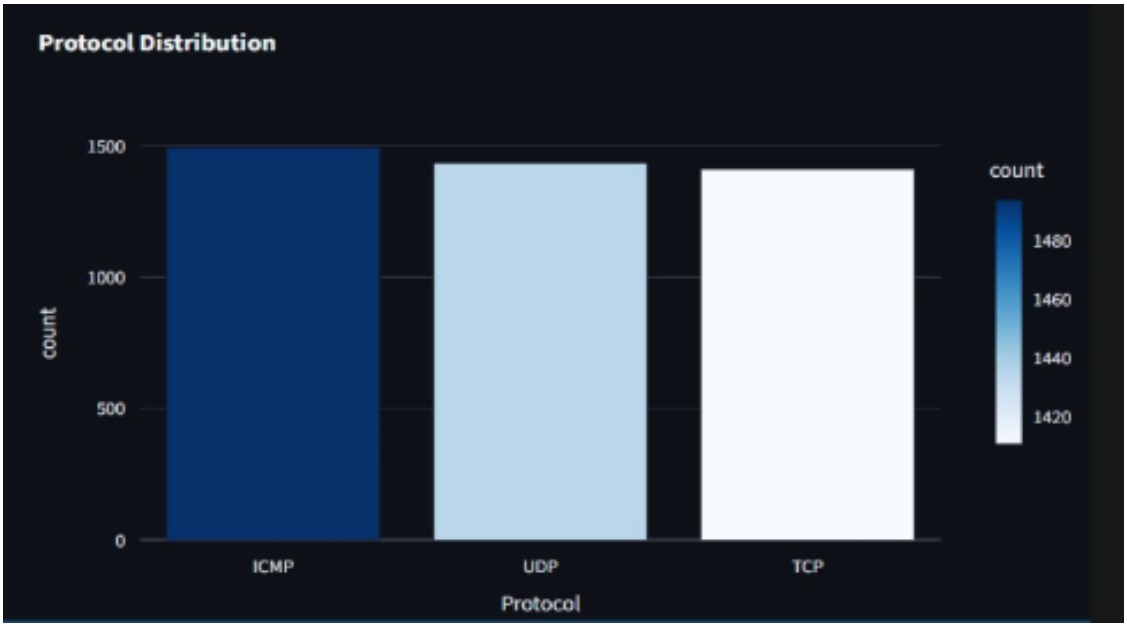


Figure 4.2.2: Protocol Distribution

This visualization presents the frequency of protocols such as TCP, UDP, and ICMP. The results indicate that certain protocols are more frequently associated with attack patterns.

Figure 4.2.3: Severity Level Breakdown

The chart illustrates the distribution of severity levels (Low, Medium, High). This helps in understanding the risk level associated with network activities.

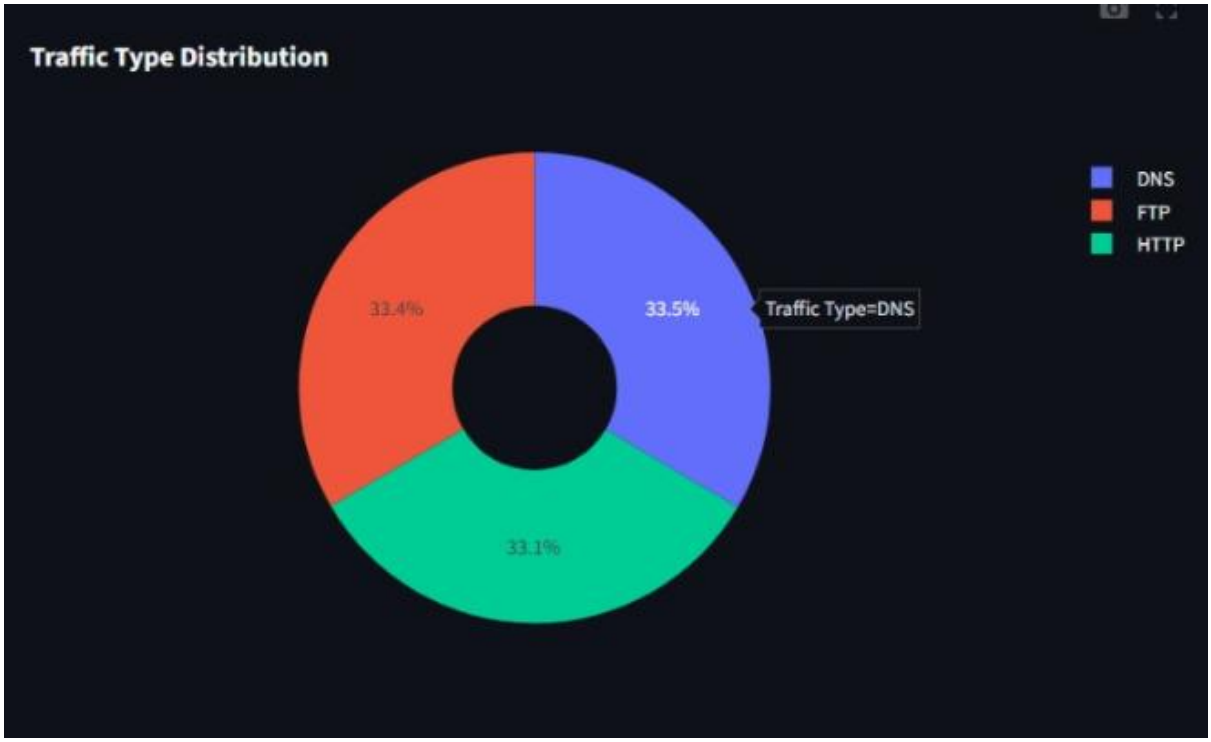


Figure 4.2.4: Traffic Type Distribution

This chart shows the distribution of traffic types (HTTP, DNS, FTP), which helps in identifying dominant traffic patterns.

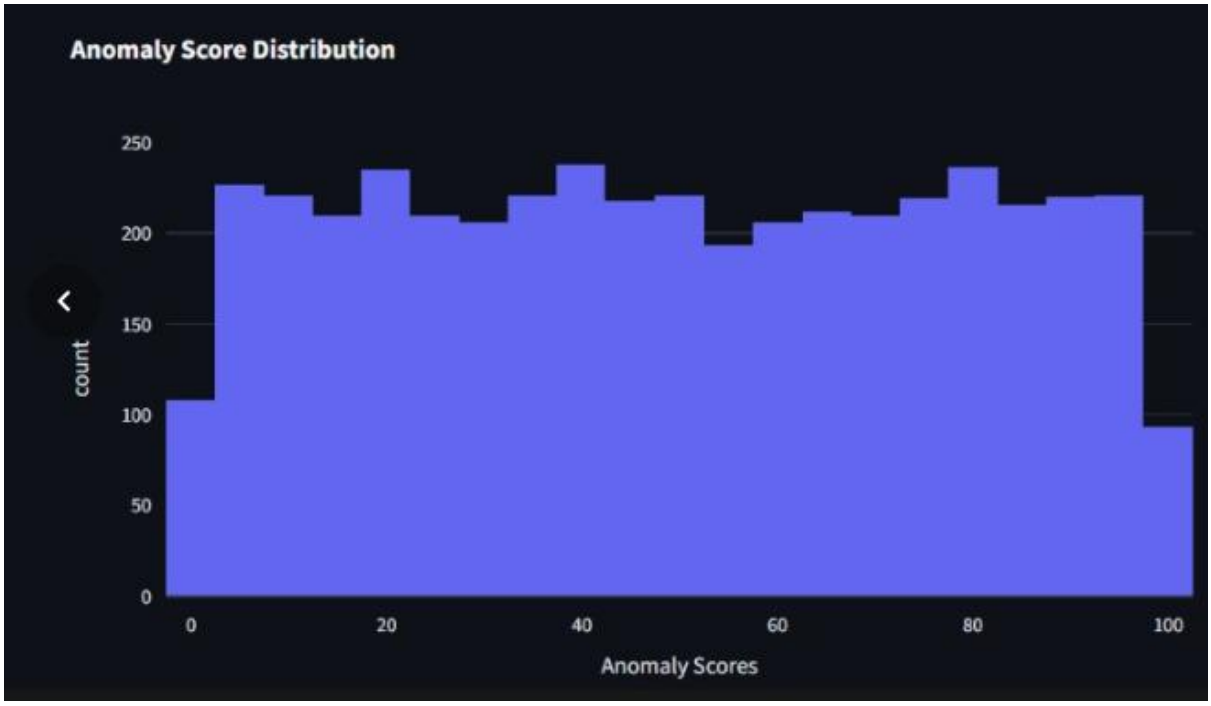


Figure 4.2.5: Anomaly Score Distribution

The histogram shows the spread of anomaly scores. With improved feature engineering, the anomaly distribution provides better separation between normal and suspicious activities.

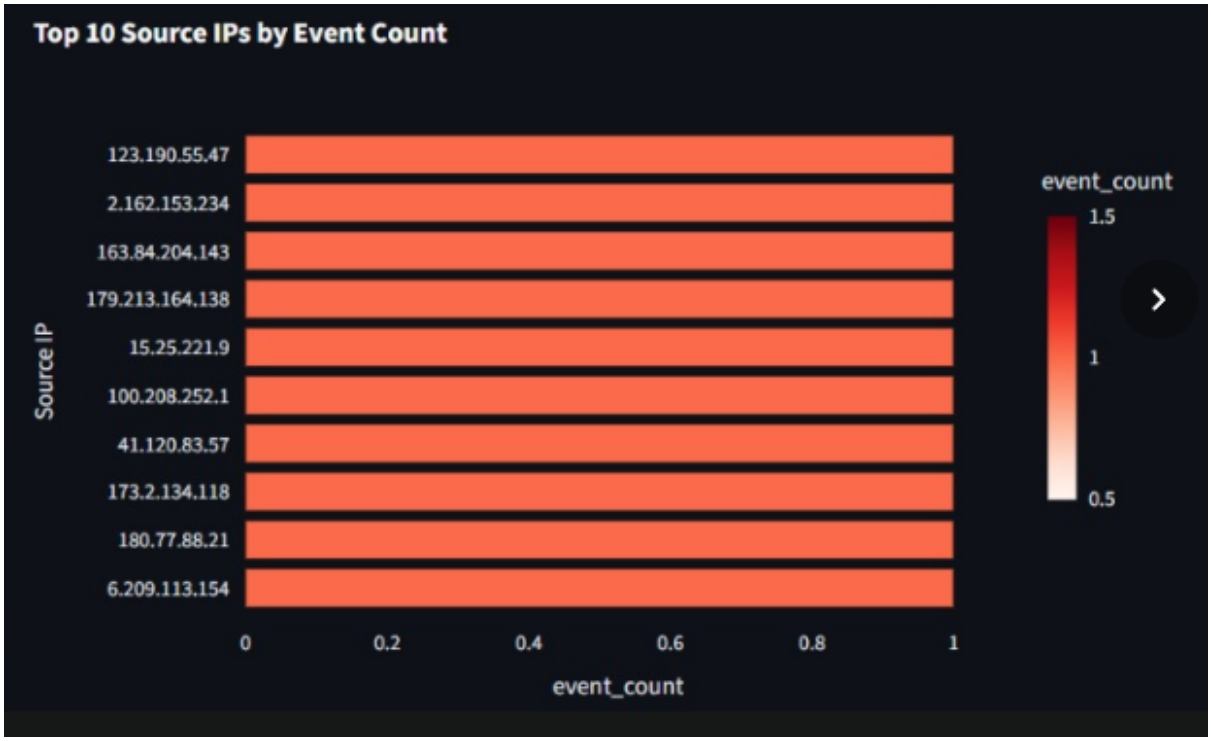


Figure 4.2.6: Top Source IP Addresses

This chart highlights the top 10 source IPs generating the highest number of events. These IPs can be considered potential sources of attacks.

Recent Activity Logs (last 15 rows)

Emit_Timestamp	Source IP Address	Attack Type	Severity Level	Protocol	Anomaly Scores	Predicted_Attack	Confidence_Pct
2026-04-03 10:11:42	91.19.10.45	Malware	Medium	TCP	29.27	Malware	74
2026-04-03 10:11:42	121.35.62.9	DDoS	High	TCP	51	DDoS	90
2026-04-03 10:11:42	172.245.170.193	Intrusion	Medium	ICMP	23.59	Intrusion	76
2026-04-03 10:11:42	99.245.144.171	DDoS	High	ICMP	31.22	Intrusion	44
2026-04-03 10:11:42	187.174.102.60	Intrusion	Low	TCP	68.01	Malware	44
2026-04-03 10:11:43	146.246.113.174	Malware	High	UDP	46.96	Malware	70
2026-04-03 10:11:43	136.244.52.250	Intrusion	Low	TCP	43.3	Intrusion	80
2026-04-03 10:11:43	148.16.26.62	Intrusion	High	TCP	58.55	Intrusion	64
2026-04-03 10:11:43	103.123.50.236	DDoS	Low	ICMP	82.31	DDoS	40
2026-04-03 10:11:43	138.147.159.209	Intrusion	Low	TCP	46.53	Intrusion	72

Figure 4.2.7: Sample Log Data

The dashboard displays recent log entries including:

- Timestamp
- Source IP Address
- Attack Type
- Severity Level
- Protocol
- Anomaly Score
- Predicted Attack
- Confidence Percentage

The dashboard updates automatically every 5 seconds, enabling real-time monitoring.

4.3 Performance Metrics

The performance of the improved model (Gradient Boosting Classifier) was evaluated using standard classification metrics.

4.3.1. Overall Performance

Metric	Value
Test Accuracy	**86.46%**
Macro Avg F1	**0.86**
Weighted Avg F1	**0.86**
Total test samples	8,000
Total training samples	32,000
Classifier	`GradientBoostingClassifier` (200 trees, depth=5)
Feature count	38

Figure 4.3.1:Model Accuracy

4.3.2. Class-wise Performance

Classification Report

Attack Type	Precision	Recall	F1-Score	Test Support
DDoS	0.87	0.89	0.88	3,039
Intrusion	0.86	0.86	0.86	2,688
Malware	0.87	0.83	0.85	2,273
Macro Average	0.86	0.86	0.86	8,000
Weighted Avg	0.86	0.86	0.86	8,000

Figure 4.3.2: Classification Report

4.3.3. Model Comparison

Model	Accuracy
Random Forest	~34.12%
Gradient Boosting	86.46%

Table 4.3:Final Accuracy Comparison

The results clearly show that the Gradient Boosting model significantly outperforms the Random Forest classifier.

4.4 Analysis

4.4.1. Interpretation of Results

The experimental results demonstrate a significant improvement in model performance after replacing the Random Forest classifier with the Gradient Boosting model. The accuracy increased from approximately 34% to 86.46%, which indicates a substantial enhancement in prediction capability.

This improvement is mainly attributed to:

- Advanced feature engineering (increase from 25 to 38 features)
- Sequential learning mechanism of Gradient Boosting
- Better handling of complex relationships in structured data

The class-wise performance shows that the model performs consistently across all attack categories. The highest recall is observed in DDoS detection (0.89), indicating strong capability in identifying such attacks. Malware detection shows slightly lower recall, suggesting that it is comparatively harder to detect.

4.4.2. Effectiveness of Feature Engineering

2. Comprehensive Feature Engineering (38 features)

Category	Features Added
Sparse null indicators	`Malware Indicators_present`, `IDS/IPS Alerts_present`, `Alerts/Warnings_present`, `Firewall Logs_present`, `Proxy Information_present`
Port range bins	`src_port_wellknown`, `dst_port_wellknown`, `dst_port_registered`, `dst_port_dynamic`, `port_diff`, `port_diff_bin`
Anomaly score bins	`anomaly_high`, `anomaly_low`, `anomaly_bin`
Packet length bins	`pkt_large`, `pkt_small`, `pkt_bin`
Timestamp features	`hour`, `dayofweek`, `month`
IP features	`src_ip_a`, `src_ip_b`, `dst_ip_a`, `dst_ip_b`
Interaction features	`anomaly_x_pkt`, `port_ratio`
Additional categoricals	`Attack Signature`, `Action Taken`, `Network Segment`, `Log Source`

Figure 4.4: Feature Engineering(38 features)

The inclusion of additional features such as anomaly score bins, port-based features, packet length categories, and interaction features significantly enhanced the model's ability to distinguish between different types of attacks.

Feature importance analysis indicates that attributes such as traffic type, protocol, action taken, and anomaly scores play a major role in prediction.

4.4.3. Comparison with Existing Methods

Compared to earlier approaches using Random Forest and traditional rule-based methods, the proposed system achieves significantly higher accuracy. The improvement demonstrates the effectiveness of combining feature engineering with advanced ensemble learning techniques.

While some existing studies report high accuracy using deep learning models, the proposed approach achieves comparable performance with lower computational complexity and faster training time.

4.4.4. Summary of Findings

- The Gradient Boosting model significantly improved accuracy to 86.46%
- Feature engineering played a critical role in enhancing performance
- The system successfully integrates Big Data processing with machine learning
- Real-time dashboard provides effective monitoring and visualization
- The proposed approach is scalable and suitable for practical applications

CHAPTER 5: CONCLUSION AND FUTURE WORK

5.1 Conclusion

This project presented a scalable and automated system for log file analysis and cyber-attack detection using Big Data and machine learning techniques. The system was designed as an end-to-end pipeline consisting of data ingestion, preprocessing, feature engineering, model building, evaluation, and real-time visualization.

Initially, a Random Forest classifier was implemented for attack detection. However, the model achieved a low accuracy of approximately 34.12%, indicating its inability to effectively capture complex patterns in the dataset. To address this limitation, the model was replaced with a Gradient Boosting classifier, which significantly improved performance.

With the introduction of advanced feature engineering, the number of features was increased from 25 to 38, incorporating additional information such as anomaly score bins, port-based features, packet characteristics, and interaction features. This enhancement, combined with the sequential learning capability of the Gradient Boosting model, resulted in a substantial increase in accuracy to **86.46%**, along with balanced precision, recall, and F1-score values of approximately 0.86.

The system also integrates PySpark for large-scale data processing and a Streamlit-based dashboard for real-time monitoring. The dashboard provides dynamic visualization of attack patterns, protocol usage, severity levels, anomaly scores, and suspicious source IP addresses, with updates occurring at regular intervals.

Overall, the project successfully demonstrates the effectiveness of combining Big Data technologies with advanced machine learning techniques to build a scalable, real-time cyber-attack detection system. The significant improvement in model performance highlights the importance of proper feature engineering and model selection in achieving accurate results.

5.2 Future Work

Although the proposed system achieves promising results, several enhancements can be made to further improve its performance and applicability:

- **Advanced Model Implementation:** Explore more powerful models such as XGBoost, LightGBM, or deep learning techniques like LSTM and neural networks to further improve detection accuracy.

- **Real-time Streaming Enhancement:** Integrate technologies such as Apache Kafka to handle high-velocity real-time data streams more efficiently.
- **Improved Feature Engineering:** Develop more domain-specific features and apply automated feature selection techniques to enhance model performance.
- **Data Quality Improvement:** Implement advanced data cleaning and normalization techniques to handle missing values and inconsistencies more effectively.
- **Hyperparameter Optimization:** Apply techniques such as Grid Search or Random Search to fine-tune model parameters for optimal performance.
- **Cloud Deployment:** Deploy the system on cloud platforms such as AWS or Azure for better scalability and real-time processing in enterprise environments.
- **Automated Alert System:** Integrate alert mechanisms to notify administrators in case of detected threats.
- **Cross-platform Compatibility:** Resolve environment-related issues (such as Hadoop configuration in Windows) to ensure smooth deployment across different systems.
- **Integration with Security Systems:** Extend the system to work with real-world security tools such as Intrusion Detection Systems (IDS) and Security Information and Event Management (SIEM) systems.

The future scope of this project lies in developing a more robust, scalable, and intelligent system capable of real-time deployment in enterprise-level cybersecurity environments.

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